CHMLTECH ONLINE JOURNAL · VOLUME 104, ISSUE 5 · RESEARCH ARTICLE

Digital Twins in Chemical Process Plants: Architecture, Modeling Approaches and Applications

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Published November 9, 2026 20 min read

ABSTRACT

Digital twins, computational representations of specific physical assets that are continuously updated with data from those assets, have attracted growing interest in the process industries, but the term is applied inconsistently and the complexity of process physics makes implementation difficult. This paper reviews the engineering basis of digital twins for chemical process plants. It traces the concept to its origin in product lifecycle management and adopts the classification into digital model, digital shadow and digital twin according to the automation of data flows between the physical and digital objects. The architecture is described in terms of the ISA-95 functional hierarchy, plant historians, OPC UA communication, edge and cloud computing, and a sequence of functional layers. Mechanistic models based on mass and energy balances, data-driven soft sensors and hybrid models are compared. Weighted least-squares data reconciliation with gross-error detection, Kalman filtering and moving-horizon estimation, and parameter estimation supported by model-based experimental design are presented as the means of keeping a twin synchronized with the plant. Applications in real-time optimization, model predictive control, fault

detection, predictive maintenance, operator training, scale-up and energy optimization are discussed, together with validation, model drift, data quality, cybersecurity under IEC 62443 and organizational adoption. A maturity framework and an illustrative reconciliation example are provided. The paper concludes that most of the value of process twins is captured by well-validated diagnostic, predictive and advisory applications before control loops are closed.

Keywords: digital twin; process systems engineering; hybrid modeling; data reconciliation; state estimation; real-time optimization; soft sensors; Industry 4.0

1. Introduction

Chemical process plants have been instrumented, simulated and computer-controlled for decades. Distributed control systems record thousands of measurements every second, steady-state flowsheet simulators are used in design and debottlenecking, and model predictive controllers run on many refinery and petrochemical units. The concept of the digital twin promises to connect these elements: a computational representation of a specific physical asset that is continuously updated with data from that asset, used to predict its behaviour, and able to inform or act on its operation. Interest in the concept has grown rapidly in manufacturing, aerospace and, more recently, the process industries [1, 2, 3].

The term is used loosely, however. Vendor literature applies it to anything from a three-dimensional plant model to a dashboard of historian trends, and the academic literature has noted that many systems described as digital twins lack an automatic data link in one or both directions [4]. For the process industries the difficulty is compounded by the complexity of the physics. Reaction kinetics, multiphase flow, phase equilibrium and fouling are difficult to model accurately, and Perno et al. suggest that this complexity partly explains why comparatively little digital-twin research has addressed the process industry [3].

This paper sets out what a digital twin of a chemical process plant is and how it is built. It covers definitions and integration levels, the information architecture, the modelling approaches (mechanistic, data-driven and hybrid), state estimation, data reconciliation and parameter estimation, the main applications, and the validation, lifecycle and organizational issues that determine whether a twin delivers value. A maturity table and an illustrative example of data reconciliation within a reactor twin are included. The emphasis is on the

engineering content of the twin, its models, data and estimators, rather than on software products.

2. Definitions and Levels of Integration

2.1 Origin of the concept

Grieves traces the digital twin to a 2002 presentation on product lifecycle management at the University of Michigan, in which a physical product, its virtual counterpart and the data and information flows connecting them were identified as the three elements of the concept [5]. Grieves and Vickers distinguish a digital twin prototype, which exists before the physical product, from a digital twin instance linked to a particular physical product throughout its life, and they emphasize the use of the twin to anticipate undesirable emergent behaviour of complex systems [5]. Tao et al. reviewed the state of the art in industry and identified the core elements as the physical entity, the virtual model, the twin data, the services built on them and the connections among them [1].

2.2 Digital model, digital shadow and digital twin

Kritzinger et al. proposed a classification based on the degree of data integration between the physical and digital objects that has become widely adopted [4]. In a *digital model*, data are exchanged manually in both directions; a conventional process simulation, tuned off-line to plant data and used for studies, falls into this category. In a *digital shadow*, data flow automatically from the physical object to the digital object, but information flows back to the plant only through human decisions; an on-line simulation that tracks the plant and advises operators is an example. In a *digital twin*, data flow automatically in both directions, so that a change in the digital object can directly change the state of the physical object [4]. Jones et al., in a systematic review, characterized the digital twin in terms of the physical entity, the virtual entity, the physical and virtual environments, the twinning process with its rate, and the physical-to-virtual and virtual-to-physical connections [2]. Figure 1 summarizes the three categories: they differ only in whether each of the two data flows is manual or automatic, not in the fidelity of the model.

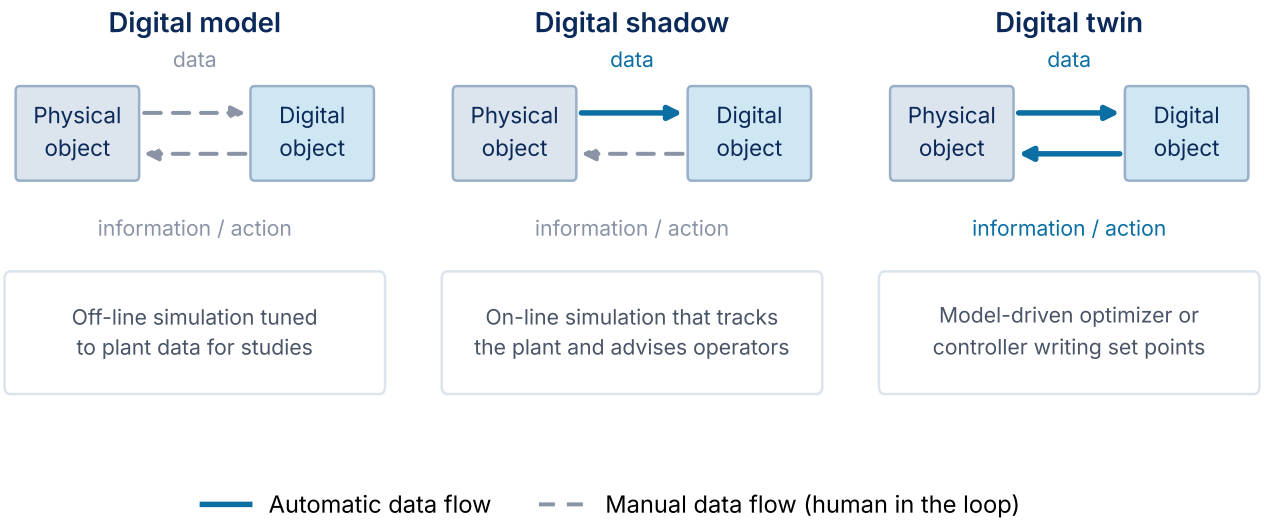


Figure 1. Digital model, digital shadow and digital twin distinguished by the automation of the data flow from the physical to the digital object and of the information flow back. Dashed grey arrows are manual flows (a human in the loop); solid blue arrows are automatic. Process examples follow Section 2.2.

Adapted from [4].

For process plants this classification has a practical consequence. Closing the loop from the digital object to the physical plant means that the twin, or a controller or optimizer driven by it, writes set points to the control system. That requirement brings the twin within the scope of process safety management, management of change and control-system cybersecurity. Many valuable process applications are therefore deliberately implemented as digital shadows, with a human in the loop, and the distinction should be made explicit in the design of the system.

> The value of a twin lies in its use, not in its label. A well-validated digital shadow that operators trust is more useful than a closed-loop twin that is switched off after its first unexplained move.

3. Architecture of a Process Plant Digital Twin

3.1 Data acquisition, historian and context

The foundation of a process twin is the plant's measurement and control infrastructure. ISA-95 (IEC 62264), which builds on the Purdue reference model, describes a functional hierarchy

of five levels: Level 0, the physical process; Level 1, sensing and manipulation through sensors, analyzers and valves; Level 2, monitoring and supervisory control by distributed control systems (DCS) and programmable logic controllers; Level 3, manufacturing operations management, including manufacturing execution systems; and Level 4, business planning and logistics, including enterprise resource planning [6]. Most process twins sit at Level 3. They draw time-series data from the plant historian, laboratory results from the laboratory information management system and equipment data from maintenance systems, and they return advice or set points to Level 2.

Measurements alone are not sufficient. A twin needs context: which tag corresponds to which stream and unit, engineering units and ranges, instrument tolerances, and the relation between tags and the model's variables. OPC Unified Architecture (OPC UA) provides a platform-independent framework for exchanging both data and information models. It supports object-oriented information modelling, client-server and publish-subscribe communication, and security functions including encryption, X.509 certificate-based authentication, message signing and audit logging [7]. Udugama et al. reviewed the role of large data sets in industrial (bio)chemical process operations and emphasized that data are useful only when they are structured, contextualized and combined with process knowledge [8].

3.2 Edge, plant and cloud computing

Computation can be placed at the edge, close to the instruments and control system, on servers within the plant network, or in the cloud. Time-critical tasks, such as state estimation for a model predictive controller or soft sensors used in control loops, are usually executed at the edge or on plant servers, where latency is low and operation does not depend on external connectivity. Computationally intensive tasks that are not time-critical, such as fleet-wide performance benchmarking, model retraining, or large what-if studies, may be executed in the cloud. The data path across these layers must be designed together with the security zones discussed in Section 7.

3.3 Functional layers

A process twin can be described as a sequence of functional layers: (i) data acquisition and validation, in which signals are checked for range, freezing, spikes and communication failures; (ii) data reconciliation and gross-error detection; (iii) state and parameter estimation, which update the model to the current condition of the plant; (iv) prediction and optimization

services; and (v) visualization and decision support or closed-loop action. Each layer depends on the quality of the one below it, which is why data validation and reconciliation, often treated as auxiliary, are central to the credibility of the whole system [9, 10]. Figure 2 shows these layers stacked on the ISA-95 levels, together with the two possible return paths to the plant: advice through operators, which keeps the system a digital shadow, and set points written directly to Level 2, which makes it a digital twin.

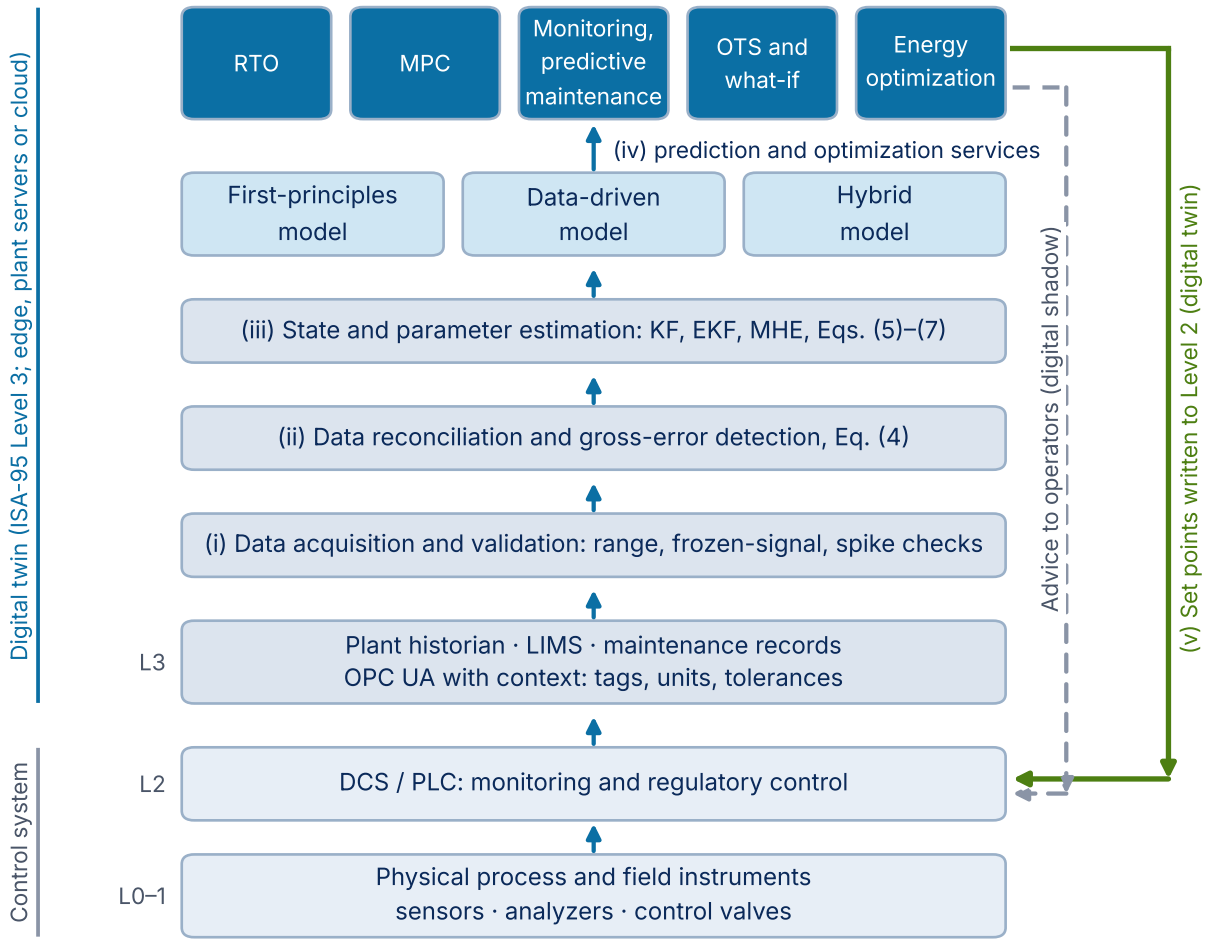


Figure 2. Layered architecture of a process plant digital twin. Plant data pass upward through the historian and OPC UA interface, validation, reconciliation and estimation to the models and applications; results return either as advice to operators (dashed) or as set points written to the control system (solid). Layer numbering (i)–(v) follows Section 3.3.

Drawn by the author from Sections 3 to 6; ISA-95 levels after [6], OPC UA after [7].

4. Modelling Approaches

4.1 First-principles models

Mechanistic models are built from conservation laws, constitutive relations and thermodynamic and kinetic data. For a continuous stirred-tank reactor (CSTR) with a single first-order exothermic reaction A → B, the component and energy balances are given by Eqs. (1) and (2).

$$\frac{dC_A}{dt} = \frac{F}{V}(C_{A,in} - C_A) - k_0 \exp\left(-\frac{E_a}{RT}\right) C_A \tag{1}$$

$$\rho\, c_p\, V \frac{dT}{dt} = \rho\, c_p\, F\,(T_{in} - T) + (-\Delta H_r)\, V\, k_0 \exp\left(-\frac{E_a}{RT}\right) C_A - UA\,(T - T_c) \tag{2}$$

Here C_A is the concentration of A, F the volumetric flow rate, V the reactor volume, k_0 and E_a the pre-exponential factor and activation energy, ρ and c_p the density and heat capacity of the mixture, ΔH_r the heat of reaction, U the overall heat-transfer coefficient, A the heat-transfer area and T_c the coolant temperature. Models of this kind extrapolate well beyond the range of the data used to calibrate them, because their structure is fixed by physics, and their parameters have physical meaning that can be checked. Their limitations are development effort, the uncertainty of some parameters (kinetics, fouling resistances, tray efficiencies) and computational cost for large plant-wide models. Commercial flowsheet simulators allow the same models to be used for design, operator training and on-line applications, which is one of the principal arguments for building process twins on mechanistic foundations. Where computational cost is limiting, surrogate models trained on the outputs of a rigorous model can be used within the twin, at the cost of an additional approximation that must itself be validated [11].

4.2 Data-driven models and soft sensors

Data-driven models learn the relation between inputs and outputs from historical data, without an explicit physical structure. Their most established use in the process industries is the soft sensor, which infers a variable that is measured infrequently or with delay, such as a product composition determined in the laboratory, from frequently measured variables such as temperatures, pressures and flows. Kadlec et al. reviewed data-driven soft sensors and identified the characteristic problems of process data that complicate them: missing values, outliers, drift, collinearity, differences in sampling rate and measurement delay. Common techniques include principal component analysis, partial least squares, artificial neural

networks and support vector machines [12]. Data-driven models are fast and can capture effects that are difficult to model mechanistically, but they are reliable only within the range of the training data, and their accuracy deteriorates when the process changes, a problem discussed in Section 7.1.

4.3 Hybrid models

Hybrid, or grey-box, models combine mechanistic and data-driven elements. Psichogios and Ungar showed in an early study that a neural network embedded in a first-principles model to estimate an unknown kinetic term could be trained with less data and extrapolated better than a purely empirical network [13]. Von Stosch et al. reviewed hybrid semi-parametric modelling in process systems engineering and distinguished serial structures, in which the data-driven part supplies parameters or terms to the mechanistic model, from parallel structures, in which it corrects the residual of the mechanistic model [14]. The two structures can be written as in Eq. (3), where f_m is the mechanistic model, g a data-driven model with weights w, and x the vector of inputs and states.

$$\text{serial: } y = f_m(x, \theta = g(x; w)); \text{ parallel: } y = f_m(x, \theta) + g(x; w) \tag{3}$$

Sansana et al. reviewed recent trends in hybrid modelling for Industry 4.0, including the ways in which process knowledge can be used to structure and constrain data-driven models [15]. Bikmukhametov and Jäschke compared approaches for combining machine learning with process physics and showed that physics-based features and hybrid structures can improve both the accuracy and the interpretability of data-driven models [16]. In a review of 50 hybrid-modelling case studies published between 2020 and 2024, Gargalo et al. concluded that hybrid models handle noisy or sparse data well and can reduce the amount of data needed, that they are a key building block for digital twins in (bio)chemical engineering, and that their adoption requires staff skilled in both process engineering and data science [17].

Table 1. Comparison of modelling approaches for process plant digital twins.

Attribute	First-principles	Data-driven	Hybrid
Basis	Conservation laws, thermodynamics, kinetics, transport	Historical input–output data	Mechanistic structure with data-driven terms or corrections
Data required	Low to moderate (for calibration)	High, covering the operating range	Moderate
Extrapolation	Good within the validity of the physics	Poor outside training data	Better than data-driven; limited by data-driven part
Development effort	High	Low to moderate	Moderate to high
Interpretability	High	Low to moderate	Moderate to high
Computational cost	Moderate to high (large flowsheets)	Low	Moderate
Typical twin uses	Real-time optimization, what-if studies, operator training, scale-up	Soft sensors, anomaly detection, short-term prediction	Soft sensors with physical constraints, kinetics with unknown terms, performance monitoring

Source: synthesis by the author based on [12, 14, 15, 16, 17].

5. Keeping the Twin Synchronized: Reconciliation and Estimation

5.1 Data reconciliation and gross-error detection

Plant measurements contain random errors and occasionally gross errors caused by instrument failure, miscalibration or leaks, and as a result they do not satisfy the mass and energy balances exactly. Data reconciliation adjusts the measurements so that they satisfy the balance constraints while staying as close as possible to the measured values, weighted by

their uncertainty [9, 18]. For steady-state linear constraints A x = 0, where y is the vector of measurements and V their error covariance matrix, the reconciled estimates $\hat{x}$ are given by the weighted least-squares problem and its analytical solution in Eq. (4).

$$\min_{\hat{x}} (y - \hat{x})^{T} V^{-1} (y - \hat{x}) \text{ subject to } A\,\hat{x} = 0;\ \hat{x} = y - V A^{T} (A V A^{T})^{-1} A\,y \tag{4}$$

The statistic $\gamma = r^{T} (A V A^{T})^{-1} r$, where $r = A y$ is the balance residual, follows a chi-square distribution with as many degrees of freedom as there are independent constraints if only random errors are present. Values above the critical value indicate a gross error, which can then be located by further tests [9]. Crowe reviewed progress in the field, including nonlinear and dynamic reconciliation, and the interaction between reconciliation and gross-error detection [18]. In a twin, reconciliation serves two purposes: it provides consistent data for model updating, and it detects instrument problems that would otherwise be absorbed into the model as spurious changes of parameters.

5.2 State estimation

Dynamic twins require estimates of states that are not measured directly, such as concentrations inside a reactor or column. The Kalman filter provides the optimal linear estimate for a linear system with Gaussian noise [19]. For a discrete-time model $x_{k} = A\,x_{k-1} + B\,u_{k-1} + w_{k-1}$ with measurement $y_{k} = H\,x_{k} + v_{k}$, where w and v have covariance matrices Q and R, the prediction and update steps are given by Eqs. (5) and (6).

$$\hat{x}_{k|k-1} = A\,\hat{x}_{k-1|k-1} + B\,u_{k-1};\ P_{k|k-1} = A\,P_{k-1|k-1}\,A^{T} + Q \tag{5}$$

$$K_{k} = P_{k|k-1}\,H^{T} (H\,P_{k|k-1}\,H^{T} + R)^{-1};\ \hat{x}_{k|k} = \hat{x}_{k|k-1} + K_{k} (y_{k} - H\,\hat{x}_{k|k-1});\ P_{k|k} = (I - K_{k}\,H)$$

For nonlinear process models such as Eqs. (1) and (2), the extended Kalman filter (EKF) linearizes the model at each step. Haseltine and Rawlings compared the EKF with moving-horizon estimation (MHE), which solves a constrained optimization over a window of past measurements, and showed examples in which the EKF converged to physically impossible estimates, such as negative concentrations, while MHE, which can enforce constraints, did not [20]. Slowly varying parameters, such as a heat-transfer coefficient that decreases with

fouling, can be estimated together with the states by augmenting the state vector with the parameter and modelling it as a random walk.

5.3 Parameter estimation and calibration

Model parameters θ are estimated by minimizing the weighted difference between measured and predicted outputs over a set of N data points, as in Eq. (7), in which Σ_i is the covariance matrix of the measurement errors.

$$\hat{\theta} = \arg\min_{\theta} \sum_{i=1}^{N} [y_i - \hat{y}_i(\theta)]^{T} \Sigma_i^{-1} [y_i - \hat{y}_i(\theta)] \tag{7}$$

Two issues are critical. First, parameters must be identifiable from the available data: routine plant data often vary too little to determine kinetic parameters independently, and correlated parameter estimates can fit historical data well while extrapolating poorly. Model-based design of experiments addresses this by choosing experimental conditions that maximize the information content for parameter estimation, a technique of particular value in pilot-plant campaigns [21]. Second, the twin must distinguish between parameters that should be updated on-line (fouling, catalyst activity, efficiency) and parameters that should remain fixed unless there is strong evidence of change (stoichiometry, thermodynamic properties), so that the estimator does not absorb measurement problems or unmodelled disturbances into physically meaningful parameters.

6. Applications

6.1 Real-time optimization and model predictive control

Real-time optimization (RTO) uses a steady-state process model, updated with plant data, to calculate set points that maximize an economic objective, typically of the form given in Eq. (8), subject to the model equations and operating constraints.

$$\max_{u} \Phi = \sum_{j} p_j F_{P,j} - \sum_{k} c_k F_{F,k} - \sum_{l} c_{U,l} U_l \text{ subject to } h(x, u, \theta) = 0,\ g(x, u, \theta) \le 0 \tag{8}$$

Here p_j and $F_{P,j}$ are product prices and flow rates, c_k and $F_{F,k}$ feed costs and flows, and $c_{U,l}$ and U_l utility costs and consumptions. Darby et al. reviewed RTO practice and identified the

main difficulties as plant–model mismatch, the need to detect steady state before updating the model, and the effort required to maintain the model over time [22]. Model predictive control (MPC), which uses a dynamic model to optimize future control moves subject to constraints, is the most widely applied advanced control technology in the process industries; the survey of Qin and Badgwell documented its industrial use and the characteristics of the commercial packages [23]. A process twin can supply both the updated steady-state model for RTO and the state estimates and model updates for MPC.

6.2 Monitoring, fault detection and predictive maintenance

A synchronized model provides a reference against which the actual behaviour of the plant can be compared. Deviations between measured and predicted values, or trends in estimated parameters such as heat-transfer coefficients, compressor efficiencies or catalyst activity, indicate faults or degradation. Venkatasubramanian et al. reviewed quantitative model-based methods for process fault detection and diagnosis, which use residuals generated by such models [24]. Predictive maintenance extends the idea to the estimation of the remaining useful life of equipment; Zonta et al. reviewed the approaches used in Industry 4.0, which range from physical degradation models to purely data-driven methods [25]. In process plants, estimated fouling resistances allow cleaning schedules to be optimized by weighing the cost of cleaning against the energy penalty and capacity loss of continued operation.

6.3 Operator training, scale-up and what-if analysis

Operator training simulators (OTS), which replicate the plant's dynamic behaviour and control system interface, are among the oldest applications of dynamic process models. Patle et al. reviewed OTS in the chemical industry and identified issues including model fidelity, the cost of development and maintenance, and the need to keep the simulator consistent with the plant as it is modified [26]. A twin that is continuously validated against the plant can reduce the last of these problems. Dynamic models are also used for what-if studies: start-up and shutdown procedures, response to equipment trips, relief scenarios and the consequences of operating changes can be examined without risk to the plant, although results used for safety decisions must be validated with the same rigour as any other engineering calculation.

Scale-up from pilot plant to commercial plant is a natural application for the process engineer. The model developed and calibrated on the pilot plant, with parameters identified by designed experiments [21], becomes the digital model of the commercial design, is used in FEED and operator training, and is then progressively converted into a digital shadow of the commercial

plant as it starts up. The twin thereby carries process knowledge across the lifecycle, rather than allowing it to be lost between the development, design and operations organizations [10].

6.4 Energy optimization

Utilities and energy are often the largest controllable operating cost. A twin of the site's steam, power and cooling systems, combined with models of the major energy consumers, allows boiler, turbine and letdown loading, and the allocation of heat recovery between units, to be optimized as prices and demands change. The same model structure as Eq. (8) applies, with utility costs and emission prices in the objective function.

7. Validation, Lifecycle Management and Challenges

7.1 Validation and model drift

A twin must be validated before it is used and monitored while it is used. Validation compares predictions with independent plant data across the intended operating range, including transients if the twin is dynamic, and establishes acceptance criteria for each output. Monitoring then tracks prediction errors, the frequency of gross-error detection, and the trends of estimated parameters. Data-driven elements face concept drift: the statistical relation between inputs and outputs changes because of catalyst ageing, feed changes, equipment modifications or seasonal effects. Gama et al. reviewed methods for detecting and adapting to drift, including windowing, forgetting and explicit change detection [27]. Kadlec et al. identified drift as one of the main reasons why soft sensors lose accuracy after deployment [12]. Every twin therefore requires an owner, a procedure for retraining or recalibration, version control of models and data, and a management-of-change process that is triggered when the plant is modified.

7.2 Data quality

Data problems are the most common practical obstacle. Instruments that are poorly calibrated, frozen or saturated, compressed historian data that hide dynamics, laboratory results that are poorly time-stamped, and missing context are all common [12, 8]. Data reconciliation exposes many of these problems [9], but it cannot compensate for unobservable variables or insufficient redundancy. Instrumentation upgrades are often a prerequisite for a useful twin and should be included in its scope and cost.

7.3 Cybersecurity

A twin connects operational technology to information technology and, often, to external networks. The ISA/IEC 62443 series defines requirements and processes for implementing and maintaining secure industrial automation and control systems throughout their lifecycle, assigns responsibilities to asset owners, product suppliers, integrators and service suppliers, and provides risk-assessment processes and security levels for system design [28]. For a twin, this implies segmenting the network so that data flow from the control system to the twin through controlled interfaces, preferring one-way data transfer where write-back is not required, using authenticated and encrypted protocols such as OPC UA with its security features enabled [7], and treating any write-back of set points as a change to the control system subject to the same security and safety review.

7.4 Organizational adoption

Perno et al. organized the enablers of and barriers to digital-twin implementation in the process industry and proposed conceptual models of their relationships; the barriers they discuss include technical complexity, data issues, cost and organizational factors [3]. Tao et al. highlighted pitfalls that limit the maturity of industrial digital twins and the challenges that remain for large-scale application [29]. In practice, the decisive factors are often a clearly defined use case with measurable benefits, the involvement of operations staff from the start, the availability of people who understand both the process and the data methods [17], and a realistic budget for maintaining the twin after the initial project is complete.

Table 2. Maturity levels for process plant digital twins.

Level	Description	Integration (after Kritzinger et al.)	Typical capability	Key requirements
1. Descriptive	Contextualized historian data, dashboards, offline models	Digital model	Visibility of current and past performance	Tag context, data quality checks
2. Diagnostic	Reconciled data, performance indicators, model-based residuals	Digital shadow	Instrument fault detection, performance gaps, fouling and efficiency tracking	Redundant measurements, reconciliation, validated models
3. Predictive	On-line state and parameter estimation, soft sensors, dynamic prediction	Digital shadow	Inferred compositions, early warning, remaining-life estimates	State estimators, drift monitoring, model governance
4. Prescriptive	Optimization of set points or schedules with operator approval	Digital shadow (human in the loop)	Advisory RTO, energy optimization, maintenance scheduling	Economic models, constraint handling, operator trust
5. Autonomous	Closed-loop optimization and control driven by the twin	Digital twin	RTO and MPC with automatic model updating	Safety and cybersecurity review, management of change, fallback modes

Source: framework proposed by the author for this paper; integration categories follow [4].

Most of the value of process twins is captured at Levels 2 to 4. Closing the loop at Level 5 is justified when the benefits are proven at Level 4 and the safety and cybersecurity implications have been reviewed.

8. Illustrative Example: Data Reconciliation in a Reactor Twin

The following example is illustrative; the plant, the measurements and their uncertainties are hypothetical. A CSTR, described by Eqs. (1) and (2), receives a feed whose flow F_1 is split downstream into a product stream F_2 and a recycle or purge stream F_3, so that the steady-state mass balance is $F_1 - F_2 - F_3 = 0$. The three mass flows are measured as 100.0, 61.0 and 41.0 t h^{-1}, each with a standard deviation of 2% of the reading. The measured balance does not close: the residual is $r = 100.0 - 61.0 - 41.0 = -2.0$ t h^{-1}. Applying Eq. (4) with $V = \mathrm{diag}(2.00^2, 1.22^2, 0.82^2)$ gives $A\,V\,A^{T} = 6.16$ and the reconciled values in Table 3. The adjustment is distributed in proportion to the variance of each measurement, so that the least precise measurement, F_1, is adjusted most. The test statistic is $\gamma = (-2.0)^2/6.16 = 0.65$, well below the 95% critical value of 3.84 for one degree of freedom, so no gross error is indicated.

In a second case, the flow transmitter on F_3 develops a bias and reads 47.0 t h^{-1}, with the other readings unchanged. The residual becomes -8.0 t h^{-1} and $\gamma = 10.0$, which exceeds the critical value and flags a gross error. If the twin had not tested for gross errors, the reconciled feed flow of 105.0 t h^{-1} would have been passed to the reactor model, and the state estimator would have interpreted the apparent increase in throughput, without a corresponding change in reactor temperature, as a loss of reaction rate or a change in heat-transfer coefficient. This is the mechanism by which instrument faults are converted into false process diagnoses, and it is why reconciliation and gross-error detection precede state and parameter estimation in the architecture of Section 3.3. Figure 3 compares the adjustments in the two cases: in case 1 every correction lies within one standard deviation of the measurement, whereas in case 2 the bias on F_3 is spread over all three streams and the feed flow is moved by 2.5 standard deviations.

Table 3. Reconciliation of the illustrative flow measurements (hypothetical data; t h^{-1}).

Stream	Measured (case 1)	σ	Reconciled (case 1)	Measured (case 2)	Reconciled (case 2, if not rejected)
F_1 (feed)	100.0	2.00	101.30	100.0	105.02
F_2 (product)	61.0	1.22	60.52	61.0	59.13
F_3 (recycle/purge)	41.0	0.82	40.78	47.0	45.89
Residual $r = F_1 - F_2 - F_3$	−2.0	—	0.00	−8.0	0.00
Test statistic γ (critical value 3.84)	0.65	—	No gross error	10.0	Gross error flagged

Source: calculation by the author using Eq. (4) and the global test described in [9]. In case 2 the standard deviation of F_3 is 0.94 t h^{-1} (2% of 47.0).

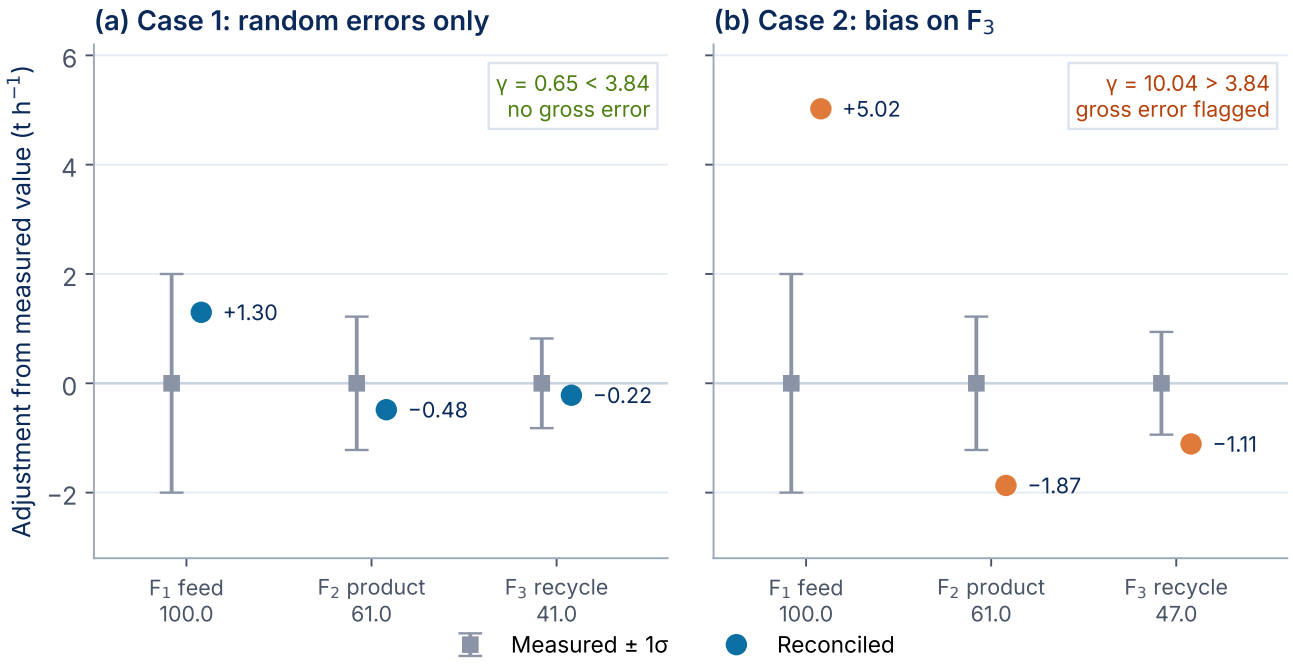


Figure 3. Adjustments made by weighted least-squares reconciliation, Eq. (4), to the three flow measurements of the illustrative example (hypothetical data). Grey bars show ±1σ of each measurement; dots show the reconciled value minus the measured value, with the measured value under each stream name. (a) Case 1, random errors only. (b) Case 2, F_3 biased to 47.0 t h^{-1}, reconciled as if the bias were not detected.

Author's calculation from Eq. (4) and Table 3.

Once validated data are available, the twin estimates the states and the slowly varying heat-transfer coefficient UA by an EKF or MHE applied to Eqs. (1) and (2), with UA appended to the state vector [20]. A sustained decline in the estimated UA at constant coolant conditions indicates fouling of the cooling surface. Because Eq. (2) shows that the heat-removal capacity limits the safe throughput of an exothermic reactor, the twin can then predict the throughput at which the reactor temperature would reach its limit, schedule cleaning by comparing the cost of the outage with the lost production, and, at Level 4 of Table 2, advise the operator of a revised feed rate.

9. Conclusion

A digital twin of a chemical process plant is not a new kind of model but a new way of using models: continuously synchronized with a specific plant through validated data, and connected to decisions about its operation. The classification into digital model, digital shadow and digital twin clarifies the degree of automation and the associated safety and

cybersecurity obligations. The architecture rests on the ISA-95 hierarchy, contextualized historian data, secure communication such as OPC UA, and a sequence of functional layers in which data validation, reconciliation and estimation precede prediction and optimization. Mechanistic models provide extrapolation and interpretability, data-driven models provide speed and flexibility within the range of the data, and hybrid models increasingly combine the two. Kalman filtering, moving-horizon estimation and weighted least-squares reconciliation keep the twin aligned with the plant, while parameter estimation, supported by designed experiments in pilot plants, provides models that can be carried from scale-up into operation. The main applications are real-time optimization, model predictive control, monitoring and predictive maintenance, operator training, energy optimization and what-if analysis. Success depends as much on data quality, drift management, cybersecurity under IEC 62443 and organizational ownership as on the sophistication of the models, and most of the value is captured by well-validated diagnostic, predictive and advisory twins before the loop is closed.

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How to cite this article

Lee, Q. F. (2026). Digital Twins in Chemical Process Plants: Architecture, Modeling Approaches and Applications. *Chmltech Online Journal*, *104*(5). https://www.chmltech.com/post/digital-twins-in-chemical-process-plants-architecture-modeling-and-applications

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